UNIVERSITÀ DI TRENTO
CiMeC Center for Mind/Brain Sciences

Master's Degree in Cognitive Science

NEURAL MASS AND SPIKING MODELS FOR THE STUDY OF NEURAL CIRCUITS

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Academic Year 2025/2026

Abstract

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Chapter 1

Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1], [2], [3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

1.1.1 Classical Stochastic Resonance

The idea of studying the effects of noise on the dynamics of a system first appeared in the work of Benzi et al. [5] and Benzi et al. [6], where the term Stochastic Resonance (SR) was used to explain the periodic behavior of earth's average temperature. A brief covering of that first SR application will be made, as it provides an intuitive understanding of the general phenomenon and a surprising link to neural systems.

The most common way to model temperature averages was using simple energy balance deterministic models:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = R_{in}(T) - R_{out}(T) \tag{1.1}$$

where C is the earth's thermal capacity, and R_{in} and R_{out} represent the total incoming and outgoing solar radiation, defined as

$$R_{in} = Q\mu(t) \tag{1.2}$$

$$R_{out} = \alpha(T)Q\mu + \varepsilon(T) \tag{1.3}$$

note that $\mu(t)$ introduces a time dependence, defined as

$$\mu(t) = 1 + \gamma\cos(\omega t) \tag{1.4}$$

where γ is a small constant, $\gamma \ll 1$. Here μ is the average incoming solar radiation oscillating due to varying orbital distances, but may represent any weak periodic forcing on the system.

In [6] a simplification is used: 3 steady-states are assumed to exist as $T_1 < T_2 < T_3$, with T_1, T_3 being stable states. 1.1 then becomes

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mu(t)\varepsilon(T)}{1+\beta(1-T/T_1)(1-T/T_2)(1-T/T_3)} - \varepsilon(T) \tag{1.5}$$

A stable steady-state is a state to which the system is attracted to, and will converge over time, in our context, it is an ”observable” climate on earth. Instead an unstable state is repelling, and thus would be at a transition point between the 2 climates.

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